

QUANG DAO

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EDUCATION

Rose-Hulman Institute of Technology

Aug 2023–May 2027

Bachelor of Science, Computer Science

GPA: 3.92/4.0

Coursework: Natural Language Processing, Deep Learning, Generative AI, Probability & Statistics, Linear Algebra

Honors: Dean's List for 9 consecutive quarters

SELECTED PRESENTATIONS AND PUBLICATIONS

Q. Dao, P. Kathalkar, & K. Eaton. "Weighted Memory Tree: Remembering What Matters for Long-Horizon LLM Agents." *Proceedings of the REALM Workshop at the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

H. Dang, **Q. Dao**, & M. Jiang. "Open, Reliable, and Collective: A Community-Driven Framework for Tool-Using AI Agents." *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP), System Demonstrations Track*, 2026.

Q. Dao, P. Kathalkar, & K. Eaton. "Weighted Memory Tree: Remembering What Matters for Long-Horizon LLM Agents." Georgia Tech Research Symposium, Atlanta, GA, USA, 2026.

Q. Dao, H. Dang, & M. Jiang. "Improving Agent Systems with Task-Specific Toolkits." Notre Dame Summer Research Symposium, South Bend, IN, USA, 2025.

EXPERIENCE

Undergraduate Research Intern | Scientific World Model for Automated Research

Summer 2026

ZLab, Princeton University | PI: Dr. Zhuang Liu

- Developed Scientific World Model with two retrieval-augmented pipelines for automated scientific research and evaluation of research proposals.
- Built an AlphaEvolve-inspired research pipeline that plans experiments, evolves research code through LLM-generated mutations and recombination, and selects candidates based on experimental comparisons with baseline implementations.
- Implemented a research evaluation pipeline that assesses novelty, feasibility, methodology, and evaluation quality using claim-conditioned literature retrieval and source-linked evidence.

Research Intern | Weighted Memory Tree for Long-Horizon LLM Agents

Spring 2026

CIPHER Lab, Georgia Tech Research Institute (GTRI) | Mentor: Kenneth Eaton

- Designed Weighted Memory Tree, organizing tasks and actions into a hierarchy with outcome-based retention scores and selection-based decay.
- Implemented utility-aware context selection with branch summarization, suppression, and reopening to control prompt size while preserving execution histories.
- Evaluated the system across three language models on GAIA-Text, improving average accuracy by 9.97 percentage points and reducing prompt-token usage by 32.8% relative to linear-history memory.

Undergraduate Research Intern | OpenTools Reasoning Agent Platform

May 2025– Aug 2025

SRP Program, DM2 Lab, University of Notre Dame | PI: Meng Jiang

- Co-developed OpenTools, a community-driven framework for standardizing, evaluating, and maintaining a library of 42 tools, distinguishing agent tool-use accuracy from intrinsic tool reliability.
- Implemented retrieval-based tool selection, refined tool metadata, and developed per-tool validation pipelines using predefined and community-contributed test cases.
- Contributed to evaluations across 15 benchmarks, demonstrating relative accuracy gains of 22.16% with GPT-4o-mini and 6.7% with GPT-5-mini over the OctoTools toolbox using the OctoTools agent framework. The full OpenTools system reduced token usage by 75% compared with OctoTools.

Senior Thesis Research | Latent Communication for Multi-Agent Systems

Fall 2026 – Present

Rose-Hulman Institute of Technology | PI: Dr. Olga Scrivner

- Develop a planner-executor architecture that transfers latent representations through a projector, enabling the receiving agent to enter at a later transformer layer.
- Implement a tool-result encoder and fusion module that combine new observations with the agent's existing latent state across tool interactions.
- Compare full-depth latent communication, early-exit models, and tool-result fusion, measuring task accuracy, tool-call correctness, computation, and latency against text-communication baselines.

Undergraduate Research Assistant | Activation Steering for LLM Agent Security

Spring 2026 – Present

Rose-Hulman Institute of Technology

- Develop inference-time activation steering with frozen model weights to defend LLM agents against prompt injection in retrieved content and tool outputs.
- Construct contrastive datasets from prompt-injection benchmarks and LLM-generated examples to derive class-specific steering vectors from safe and attack-following hidden-state representations.
- Investigate steering-vector construction and intervention placement across model layers, comparing targeted and random placements against an unsteered baseline to assess attack resistance and task performance.
- Design a reinforcement-learning controller that uses hidden-state risk estimates to adjust steering strength, with rewards balancing defense success and legitimate task performance.

Undergraduate Research Assistant | Computational Tools for Chemistry

Feb 2025–Feb 2026

Mine Lab, University of Notre Dame | PI: Xiangliang Zhang

South Bend, IN

- Authored the “Yield Prediction” and “LLMs for Chemistry” chapters in the open-source *Data-Chemist Handbook*
- Contributed to the open-source *Data-Chemist Handbook* applying machine-learning and deep-learning techniques for chemical reaction prediction, process optimization, and yield forecasting, which is used as instructional material in Notre Dame's Chemistry and CS summer camp for high school students
- Designed hands-on tutorials and end-to-end yield-prediction notebooks using the Buchwald–Hartwig reaction dataset, covering math foundations and training pipelines for RNNs, GNNs, Random Forests, and Neural Networks, improving yield-prediction R^2 relative to baseline models

Machine Learning Intern

Dec 2024–May 2025

Rose-Hulman Ventures

Terre Haute, IN

- Fine-tuned and evaluated a generative adversarial network (GAN) in a three-person team to automatically pinpoint surgical-instrument tips and critical organ landmarks in live laparoscopic video
- Built an interactive GUI that streamed laparoscopic video and visualized tracked instrument trajectories on STL-based 3D anatomy.

Research Assistant | Computer Science Education Research

Apr 2024–May 2025

Rose-Hulman Institute of Technology | PI: Joe Hollingsworth

Terre Haute, IN

- Instrumented student code with an AspectJ tracer to record execution traces and errors across 2,000+ submissions.
- Clustered student error patterns with NumPy, visualized recurring mistakes with Matplotlib, and revised test suites, reducing recurring student errors by 30%.
- Integrated the Gemini API to detect when students struggle with the same errors and deliver targeted conceptual hints, guiding independent debugging without revealing full solutions

Software Engineer Intern

Jun 2024–Aug 2024

TAPTAP

Ho Chi Minh City, Vietnam

- Developed and maintained an internal analytics dashboard to monitor 10,000+ customer voucher redemptions
- Implemented a custom NestJS decorator for role-based access control and audit logging, preventing 50+ unauthorized access attempts
- Built a Telegram bot that checked voucher-redemption data in MongoDB hourly and sent alerts when configured thresholds were exceeded.
- Optimized frontend performance with SWR and Redis caching, reducing redundant API calls and cutting page load time to under one second

PROJECTS

- LLM Tool-Use Error Classification | *Python, PyTorch, Transformers, HuggingFace*

Sep 2025

 - Fine-tuned a Llama 3.2 3B model with a classification head and gradual unfreezing to detect common LLM tool-use errors, improving classification accuracy compared to a zero-shot baseline
 - Used an 8-bit optimizer and bfloat16 precision to reduce GPU memory requirements during training.
 - Analyzed error distributions to identify dominant tool-use failure modes, providing signals to refine agent behavior and prompt design
- Image Stitching Panorama | *Python, NumPy*

March 2025

 - Built a panorama-stitching pipeline using vectorized Harris corner detection, MOPS descriptors, SSD-based feature matching, RANSAC-based homography estimation, and image warping and blending.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, TypeScript
Libraries/Frameworks: PyTorch, LangChain, HuggingFace, NumPy, TensorFlow, scikit-learn, Matplotlib, SentenceTransformers, React, Node.js, NestJS
Data Storage/Developer Tools: MongoDB, PostgreSQL, Microsoft SQL Server, Git, Docker

TEACHING AND MENTORING EXPERIENCE

- Systems Programming Teaching Assistant

Nov 2025–May 2026

 - Supported weekly lab sections for a C-based systems programming course, guiding students through debugging pointer-heavy code, dynamic memory management and file I/O
 - Answered conceptual and implementation questions during lab sessions on topics such as compilation and memory
- Math Teaching Assistant

Sep 2025 – Present

 - Hosted weekly tutoring sessions to clarify core topics and support student learning
 - Created review sheets and practice sets; graded assignments with clear rubrics
- AskRose Math and Science Tutor

Aug 2024

 - Provided one-on-one tutoring to middle and high school students in mathematics and science through phone and online sessions
 - Assisted students in subjects including algebra, calculus, physics, and chemistry, ensuring clarity of concepts and improving problem-solving skills